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ORIGINAL ARTICLE

Relationship between recovery of neuromuscular function and subsequent capacity to work above critical power

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Abstract

Purpose To investigate the relationship between the recovery of neuromuscular fatigue and the recovery of amount of work done above critical power (*W´*).

Methods Ten healthy men performed, on different days, constant work rate exercises until task failure to determine critical power (CP) and *W´*. In the three following visits, participants performed two exhausting constant work rate exercises estimated to induce task failure within 6 min ($P6_1$ and $P6_2$), interspaced by 3, 6 or 15 min of recovery. Neuromuscular function was assessed before and periodically after the $P6_1$ using percutaneous electrical femoral nerve stimulation. The *W´* recovery was measured from the total work performed above CP during the $P6_2$.

Results The $P6_1$ induced a full use of *W´* and a reduction in maximal voluntary contraction (MVC, $-19 \pm 4\%$), voluntary activation (VA, $-6 \pm 2\%$) and twitch force stimulated at 1 Hz ($-37 \pm 11\%$), 10 Hz ($-50 \pm 16\%$) and 100 Hz ($-32 \pm 11\%$), when compared to baseline ($P < 0.05$). The time constant of VA recovery was significantly faster than the time constant of *W´* recovery ($P < 0.05$), but there was no significant difference between the time constant of *W´* recovery and the time constant of recovery of MVC or twitch force stimulated at 1, 10 and 100 Hz ($P > 0.05$). However, the time constant of *W´* recovery was only associated to the time constant of MVC recovery ($r = 0.73$, $P < 0.05$).

Conclusion The *W´* recovery is not associated to the recovery of peripheral or central fatigue alone. Rather, *W´* seems to be associated to the recovery of the overall capacity to generate force.

Keywords Exercise tolerance · Neuromuscular fatigue · Neuromuscular recovery · Power–time relationship · *W´*

Abbreviations

Δ	Work rate difference between GET and the $\dot{V}O_{2max}$
AV	Voluntary activation
ANOVA	Analysis of variance
Ca^{2+}	Calcium
CI	Confidence interval
CNS	Central nervous system
CP	Critical power
EMG	Surface electromyography
ES	Electrical stimulation
GET	Gas exchange threshold
HR	Heart rate
MVC	Maximal isometric voluntary contraction
M-wave	Muscle action potential
PCr	Phosphocreatine
P6	Exercise intensity estimated to induce task failure within 6 minutes
$Q_{tw,unpot}$	*Unpotentiated* twitch torque
$Q_{tw,pot}$	Quadriceps potentiated twitch evoked by single pulse
Q_{tw10}	Quadriceps potentiated twitch evoked by paired pulse at 10 Hz

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Q_{tw100}	Quadriceps potentiated twitch evoked by paired pulse at 100 Hz
RPE	Rating of perceived exertion
$t_{1/2}$	Half-time to recovery
t	Time to task failure
$\dot{V}CO_2$	Carbon dioxide production
$\dot{V}E$	Minute ventilation
$\dot{V}O_{2max}$	Maximal oxygen uptake
W'	Amount of work done above critical power

Introduction

Exercise-induced neuromuscular fatigue can be attributed to several processes along the motor pathway, resulting in a reduction in muscle force (Gandevia 2001). When the decline in muscle force originates from processes residing within the central nervous system (CNS), it is named "central fatigue" (Gandevia 2001). On the other hand, "peripheral fatigue" attributes the decline in muscle force to processes residing in motoneurons, neuromuscular junction, and/or within skeletal muscle fibers (Fitts 1994). Central fatigue is commonly assessed by changes in supramaximally stimulating the peripheral motor nerve during an isometric maximum voluntary contraction (MVC), while peripheral fatigue is assessed by changes in supramaximally stimulating the peripheral motor nerve immediately after the MVC with relaxed muscle (i.e., potentiated twitch force) (Merton 1954; Behm et al. 1996).

The origin of exercise-induced fatigue is, however, dependent on the exercise intensity (Valentini and Nelson 1985; Enoka and Stuart 1992). Exercise performed in the heavy-intensity exercise domain [i.e., below the asymptote of the power–time hyperbola (critical power, CP) but above the gas exchange threshold (GET)] will result in both central and peripheral fatigue (for review, see Burnley and Jones 2016). On the other hand, compared to the heavy-intensity exercise domain, exercise performed in the severe-intensity exercise domain (i.e., above CP) will develop considerably more peripheral fatigue and less central fatigue (Burnley and Jones 2016). Task failure during severe-intensity exercise will coincide with the depletion of the curvature constant of the power–time hyperbola (i.e., W', which represents the finite amount of work in excess of CP) (Moritani et al. 1981; Black et al. 2017). Some evidence suggest that W' may be constrained by the magnitude of peripheral fatigue accrued during a severe-intensity exercise (Broxterman et al. 2015; Schafer et al. 2019; Zarzissi et al. 2019).

The recovery of W' and its relation to the recovery of peripheral fatigue is, however, underexplored. An understanding of the factors accountable for recovery between sets of exercise performed above CP is important as this is common in many sports (e.g., soccer). The time course of W' recovery has a curvilinear behavior, with a fast recovery within the first few seconds after the exercise, followed by a slow recovery during the following minutes (Skiba et al. 2014; Broxterman et al. 2016). The estimated half-time ($t_{1/2}$) of W' recovery is ~ 4 min (Ferguson et al. 2010; Skiba et al. 2015) and the capacity to tolerate subsequent exercise above CP is associated to the recovery of W' (Coats et al. 2003; Ferguson et al. 2010). On the other hand, the recovery of potentiated twitch force (i.e., peripheral fatigue) seems to be much longer (Tupling et al. 2000; Hureau et al. 2016). Recovery of peripheral fatigue may be incomplete for some hours due to prolonged impairment in intracellular Ca^{2+} release or sensitivity (Carroll et al. 2017). Therefore, whether the relationship between W' and the peripheral fatigue after a severe-intensity exercise is maintained during the recovery is unknown and deserves further investigation.

In the present study, we investigated the relationship between the time course of W' recovery and neuromuscular function (central and peripheral fatigue) after an exhausting, constant work rate exercise performed above CP (i.e., severe-intensity exercise). We hypothesized that the W' recovery would be faster and not related to the recovery of peripheral fatigue.

Material and methods

Participants

Ten healthy men participated as volunteers (age: 24 ± 4 years, height: 176 ± 5 cm, body mass: 75 ± 9 kg). Participants were classified as physically active in accordance with the International Physical Activity Questionnaire (Craig et al. 2003). A previous study showed that repeated-measures ANOVA detected significant difference in W' recovery between 2, 6 and 15 min of recovery after a supra-CP conditioning bout (Ferguson et al. 2010). From the mean and standard deviation reported in that study, we estimated a f effect for repeated-measures ANOVA of 4.0. Assuming an alpha error of 0.05 and a beta error of 0.95, the calculated sample size necessary to detect difference in W' between the time points of recovery was six participants. However, the starting sample size was increased to ten participants to account for not complying with instructions or drop out during the data collection. The required sample size was estimated using G*Power software (Heinrich-Heine-University Düsseldorf, version 3.1.9.4, Düsseldorf, Germany). Participants signed a written informed consent before starting the trials. The study was conducted according to the Declaration of Helsinki and approved by the Research Ethics Committee of the Federal University of Pernambuco.

Experimental design

Participants visited the laboratory on seven different occasions and trials were performed within 3 weeks and at least 48 h apart. In the first visit, a maximal incremental test was performed on a cycle ergometer (Ergo-Fit 167, Pirmasens, Germany) to determine GET, maximal aerobic work rate and maximal oxygen uptake ($\dot{V}O_{2max}$). In the next three visits, constant work rate trials were performed until task failure to determine the CP and *W´*. After these trials, participants were familiarized with the neuromuscular function procedures. In the three last visits, participants performed two constant work rate trials at an exercise intensity estimated to induce task failure within 6 minutes ($P6_1$ and $P6_2$). The $P6_1$ and $P6_2$ were interspaced by 3, 6 or 15 min of passive recovery. The neuromuscular function was assessed: (1) before and at 1 and 2 min after the $P6_1$ for the 3-min recovery trial; (2) before and at 5 min after the $P6_1$ for the 6-min recovery trial; (3) before and at 14 min after the $P6_1$ for the 15-min recovery trial and; (4) at 1 min after the $P6_2$. The validity of $\dot{V}O_{2max}$ measurement was assessed by comparing the $\dot{V}O_{2max}$ values obtained in the maximal incremental test with the values obtained at the end of $P6_1$ and $P6_2$ trials (Poole and Jones 2017).

The seat position of the cycle ergometer was recorded in the first trial and replicated during the subsequent trials. All trials were performed at the same time of day to avoid any effect of circadian rhythm. Participants registered food and beverages consumed during the 24 h preceding the first trial and repeated this diet regime before the subsequent trials. They were instructed to refrain from exercise and not to consume food and beverages containing caffeine 24 h before trials. All tests were performed at least 2 h postprandial.

Maximal incremental test

The maximal incremental test started with a 5-min warm-up at 70 W and then the work rate was increased 30 W every 3 min until task failure (Felippe et al. 2018). Participants were asked to maintain pedal cadence between 70 and 80 rpm throughout the test; task failure was assumed when the participants were unable to maintain this target cadence. Oxygen uptake ($\dot{V}O_2$), carbon dioxide production ($\dot{V}CO_2$), and ventilation ($\dot{V}E$) were measured breath-by-breath using an automatic metabolic cart (Cortex, Metalyzer 3B, Leipzig, Germany), which was calibrated before each test using a 3-L syringe and a standard gas of known O_2 (12%) and CO_2 (5%) concentrations. Heart rate (HR) was measured using a heart monitor transmitter (Polar, Kempele, Finland) connected to the metabolic cart. The $\dot{V}O_{2max}$ was identified as the highest 30-s $\dot{V}O_2$ values during the test. Two experienced investigators identified the GET from a first disproportionate increase in $\dot{V}CO_2$ relative to $\dot{V}O_2$, an increase in $\dot{V}E/\dot{V}O_2$ without an increase in $\dot{V}E/\dot{V}CO_2$, and an increase in end-tidal O_2 pressure with no fall in end-tidal CO_2 pressure (Meyer et al. 2005).

CP and *W´* determination

Participants performed a 5-min warm-up at 100 W, followed by a 5-min rest, and then the work rate was adjusted to 70% of the difference between the GET and $\dot{V}O_{2max}$ (Δ70), 80% of the difference between the GET and $\dot{V}O_{2max}$ (Δ80), or 100% $\dot{V}O_{max}$. The trials were performed on different days with the order determined using a balanced crossover Latin square design. Participants cycled until task failure, which was defined as an inability to maintain pedal cadence between 70 and 80 rpm. Time to task failure was measured to the nearest second. Participants were unaware of work rate or elapsed time during the trials. The exercise intensities were chosen to yield time to task failure between 2 and 15 min (Jones et al. 2008). Strong verbal encouragement was provided throughout the trials.

Individual CP and *W´* were estimated from the work rates and the corresponding time to task failure rates, using the CP linear inverse of time (Arcoverde et al. 2017; Vanhatalo et al. 2007):

$$\text{Work rate} = \left(W'/t\right) + \text{CP} \tag{1}$$

where *W´* is the estimated constant curvature of hyperbole power–time, *t* is the time to task failure, and CP is the estimated critical power.

The individual P6 work rate was estimated by solving Eq. (1), as previously determined (Ferguson et al. 2010):

$$\text{P6} = \left(W'/360\right) + \text{CP} \tag{2}$$

The power–time integral above CP during $P6_1$ was used to determine the total work performed above CP. Because the exercise was performed until task failure at $P6_1$, which is expected to fully deplete the *W´*, the power–time integral above CP during $P6_2$ was used to determine the magnitude of recovery of *W´* (Ferguson et al. 2010).

Experimental trials

Participants arrived at the laboratory, rested for 5 min and then performed a 5-min warm-up cycling at 70 W. They then performed a specific warm up on a custom-made bench chair (two 5-s isometric contractions at 80 and 100% of the maximal voluntary isometric contraction recorded during the familiarization sessions, interspersed by a 30-s recovery). Thereafter, three maximal voluntary isometric contractions (MVC) with electrical stimulation of the femoral nerve (ES) were performed for assessment of neuromuscular function at baseline (30-s recovery between the MVCs). Participants then cycled for 5 min at 70 W, rested for further 5 min and performed the $P6_1$ trial until task failure. After the $P6_1$ trial, participants recovered for 3, 6 or 15 min, with neuromuscular function (one MVC + ES) assessment at 1 and 2 min (for the 3-min recovery trial), at 5 min (for the 6-min recovery trial) and at 14 min (for the 15-min recovery trial). Thereafter, participants returned to the cycle ergometer and cycled until task failure at the same exercise intensity used in $P6_1$ ($P6_2$). Neuromuscular function (one MVC + ES) was reassessed at 1 min post-$P6_2$ trial. Each recovery time was tested on separate days, with the order of recovery time being determined using a balanced crossover Latin square design.

During both $P6_1$ and $P6_2$, $\dot{V}E$, $\dot{V}O_2$, $\dot{V}CO_2$ and HR were continually measured throughout exercise and the mean value of the last 30 s used for further analysis.

Neuromuscular function

The evoked force and EMG response of the right vastus lateralis during and after MVC was used to assess neuromuscular function (Merton 1954). The participants seated on a custom-made bench chair, with hip joint angle set at 120° and the knee joint angle set at 90°. A non-compliant cuff attached to a calibrated linear strain gauge (EMG System of Brazil, São Jose dos Campos, Brazil) was fixed to the right ankle just superior to the malleoli for force measurement. A monopolar 0.5-cm diameter cathode electrode was positioned at the right femoral nerve and an anode was positioned on the gluteal fold opposite the cathode. A constant-current electrical stimulator (Neuro-TES; Neurosoft, Ivanovo, Russia) was used to deliver single and paired electrical stimulus on femoral nerve.

During the familiarization sessions, the optimal intensity of stimulation was determined by increasing intensity of a single electrical stimulus (1 Hz and 80 µs duration) by 30 V every 30 s until a plateau occurred in the *unpotentiated* twitch torque ($Q_{tw,unpot}$) and peak-to-peak amplitude of the compound muscle action potential (M-wave) (Skurvydas et al. 2010). The plateau was identified when no further increase in $Q_{tw,unpot}$ and M-wave amplitudes were observed despite three consecutive increases in stimulation intensity. The position of electrodes was marked with indelible ink to ensure identical placement and the plateau for both $Q_{tw,unpot}$ and M-wave was double checked before each trial.

During each MVC, participants were instructed to produce their maximal torque and maintain for 5 s. Visual feedback of instantaneous torque was provided. Square wave stimuli (80 µs) were delivered to the femoral nerve with the stimulation intensity set at 120% of the plateau (270 ± 15 V). The first stimulus was applied during the MVC as soon the plateau in isometric force had been reached (*superimposed twitch*). Potentiated twitch torque evoked by single ($Q_{tw,pot}$), and paired pulses at 10 Hz (Q_{tw10}) and 100 Hz (Q_{tw100}) were measured 2, 4 and 6 s after each MVC, respectively (Fig. 1).

The MVC was measured as the greatest force attained prior to the twitch onset torque (250 ms window) and used as a marker of global fatigue. The $Q_{tw,pot}$, Q_{tw10} and Q_{tw100} were measured as the peak of twitch torque generated for the corresponding stimulus and used as markers of peripheral fatigue (Millet et al. 2012). As a marker of central fatigue (Millet et al. 2012), the maximal voluntary activation (VA) was measured by the interpolated twitch technique, using the following equation (Merton 1954):

$$1 - \left(\text{superimposed twitch}/Q_{tw,pot}\right) \cdot 100 \tag{3}$$

where superimposed twitch is the difference between twitch at onset torque and peak twitch torque during the MVC.

The EMG signal was recorded with a sample rate of 2000 Hz by a 16-bit A/D converter ((EMG System of Brazil, São Jose dos Campos, Brazil). The EMG signal was amplified with octal bio-amplifier with a bandwidth frequency ranging from 20 to 500 Hz (input impedance = 109 Ω, common mode rejection ratio = 9 100 dB, gain = 2000), transmitted to the computer and further analyzed in Matlab (version R2013a, Mathworks Inc.). The M-wave peak-to-peak amplitude was measured for each single stimulus of 1 Hz. The beginning of the M-wave was considered when there was an increase of 2 SD above the baseline EMG signal and the ending when the EMG signal returned to less than 2 SD of the baseline (Rodriguez-Falces and Place 2016).

Analysis

The baseline for the neuromuscular parameters was assumed as the average between the three pre-$P6_1$

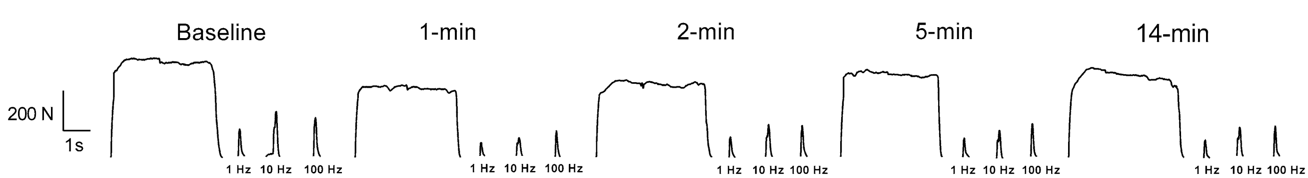

Fig. 1 Typical raw trace of maximal voluntary contraction with superimposed twitch, potentiated quadriceps twitch torque evoked by single ($Q_{tw,pot}$) and paired pulses of 10 Hz (Q_{tw10}) and 100 Hz (Q_{tw100})

measurements. The MVC and neuromuscular parameters measured at 1, 2, 5 and 14 min after the $P6_1$ were used to determine the time course of recovery of neuromuscular function. The W' and neuromuscular parameters were plotted against the recovery time and fitted using a single-exponential model:

$$y = \text{ baseline } + A\left[1 - e^{-(t-\delta/k)}\right] \tag{4}$$

where y is W' or a given neuromuscular parameter at a given time, baseline is the W' or a given neuromuscular parameter immediately after $P6_1$ (i.e., 1-min post-exercise measurement), A is the asymptotic amplitude, t is the time, δ is the time delay and k is the time constant.

The time constant was used to represent the speed with which W' or neuromuscular parameters were recovered. Neuromuscular parameters assessed 1 min after the $P6_2$ were used to determine the level of $P6_2$-induced neuromuscular fatigue (i.e., baseline to post-$P6_2$ changes and pre- to post-$P6_2$ changes).

Statistical analysis

The normal data distribution was confirmed by the Shapiro–Wilk test. The reliability of the assessment of neuromuscular function was obtained by calculating the within-subject coefficient of variation between the second and third familiarization sessions. One-way repeated-measures ANOVA was used to compare: (1) the total work performed above CP during the three $P6_1$ and the W' determined by the linear inverse of time model; (2) the $\dot{V}O_2$ reached during $P6_1$ and $P6_2$ and $\dot{V}O_{2max}$ achieved during the incremental test; (3) the time course of neuromuscular fatigue (baseline, and 1, 2, 5 and 14 min post-exercise); and (4) the time constant of the parameters. The time to task failure, the amount of work performed above CP and the end systemic responses between $P6_1$ and $P6_2$ were compared using two-way repeated-measures ANOVA, having trial ($P6_1$ and $P6_2$) and recovery time (3, 6 and 15 min) as factors. When necessary, the Tukey post hoc test was used to locate the differences. Pearson coefficient of correlation was used to determine the association between the time constant of W' and the time constant of the parameters of neuromuscular fatigue and association between neuromuscular fatigue after $P6_1$ and $P6_2$ and W'. To check a possible training effect due to several successive exercise sessions, a paired t test was used to compare the time to task failure in $P6_1$ between the first and last experimental trial. Statistical significance was reported when $P < 0.05$. Analyses were carried out using Statistics Package for Windows (version 10; StataSoft, Tulsa, OK). Data are reported as mean ± SD.

Results

The GET was identified at 142 ± 15 W. The maximal work rate was 234 ± 36 W and $\dot{V}O_{2max}$ 2.8 ± 0.5 L min^{-1} (38 ± 6 mL·kg^{-1} min^{-1}). The maximal RPE was 19 ± 1. The mean work rate corresponding to Δ70, Δ80%, and 100% $\dot{V}O_{2max}$ was 202 ± 23 W, 216 ± 29 W, and 234 ± 33 W, respectively. The corresponding times to task failure were 726 ± 179 s, 570 ± 188 s, and 393 ± 119 s, respectively. The estimated CP and W' were 173 ± 29 W (95% CI 85–260 W) and 22 ± 10 kJ (95% CI 9–35 kJ), respectively, with the standard error of estimate for CP and W' of 7.0 ± 5.0 W and 1.8 ± 1.0 kJ, respectively. The P6 work rate was 223 ± 8 W. The within-subject coefficient of variation were: work measured above CP during the three $P6_1$ trials (i.e., an estimate of W' variation) = 6.1 ± 3.4% (range 1.3–9.1%), MVC = 2.1 ± 1.8% (range 1.2–4.6%), VA = 1.6 ± 1.0% (range 0.6–2.1%), $Q_{tw,pot}$ = 4.7 ± 3.1% (range 2.2–6.6%), Q_{tw10} = 5.2 ± 4.1% (range 1.1–7.2%) and Q_{tw100} = 5.8 ± 3.8% (range 2.3–6.7%).

Time to task failure and systemic responses to $P6_1$ and $P6_2$

As expected, time to task failure was not different ($P > 0.05$) between the three $P6_1$ (Table 1). However, time to task failure was lower in all $P6_2$ compared to all $P6_1$ ($P < 0.05$). In addition, time to task failure at $P6_2$ was longer after 15 min of recovery, compared to both 3 ($P = 0.01$) and 6 min of recovery ($P = 0.01$). There was no significant difference between the 3- and 6-min recovery trials ($P = 0.06$).

The end $\dot{V}O_2$ was not different between trials (Table 1) and not different from the values achieved during the maximal incremental test ($P = 0.45$). The end $\dot{V}E$, $\dot{V}CO_2$, HR, and RPE were also not different among the trials (all $P > 0.05$).

Amount of work performed above CP

The amount of work performed above CP was not different among the three $P6_1$ trials and not different from W' obtained from the inverse time model ($P = 0.18$). However, the amount of work performed above CP was lower in all $P6_2$ compared to all $P6_1$, but values were significantly higher after 15 min of recovery compared to both 3 ($P = 0.01$) and 6 min of recovery ($P = 0.01$), without significant differences between the 3- and 6-min recovery trials ($P = 0.08$, Table 1). Consequently, the reconstitution of W' was significantly higher after 15 min of recovery than after 3 or 6 min of recovery (Fig. 2a).

Table 1 Time to task failure, work performed above critical power and systemic responses during the $P6_1$ and the $P6_2$ performed 3, 6 and 15 min after the $P6_1$

	$P6_1$			$P6_2$		
	3 min	6 min	15 min	3 min	6 min	15 min
Time to task failure (s)	377 ± 61	397 ± 47	400 ± 61	$142 \pm 26^{\dagger}$	$180 \pm 18^{\dagger}$	$254 \pm 32^{\dagger\ddagger}$
Work above CP (kJ)	21 ± 9	23 ± 8	23 ± 9	$8 \pm 4^{\dagger}$	$9 \pm 3^{\dagger}$	$13 \pm 5^{\dagger\ddagger}$
$\dot{V}E$ (L min^{-1})	128 ± 18	132 ± 20	130 ± 19	123 ± 19	127 ± 21	129 ± 18
$\dot{V}O_2$ (L min^{-1})	2.8 ± 0.4	2.9 ± 0.5	2.9 ± 0.5	2.7 ± 0.4	2.8 ± 0.4	2.8 ± 0.5
$\dot{V}CO_2$ (L min^{-1})	3.5 ± 0.6	3.6 ± 0.5	3.5 ± 0.5	3.2 ± 0.3	3.3 ± 0.5	3.3 ± 0.8
HR (beats min^{-1})	182 ± 9	182 ± 8	182 ± 10	181 ± 17	181 ± 12	180 ± 12
RPE_{final} (unit)	19 ± 2	20 ± 1	19 ± 1	19 ± 1	19 ± 1	19 ± 1

Values are expressed as mean ± SD ($n = 10$)

CP, critical power; $\dot{V}E$, ventilation; VO_2, oxygen uptake; VCO_2, carbon dioxide production; HR, heart rate; RPE_{final} rating of perceived exertion at the task failure

†Significantly lower than $P6_1$ ($P < 0.05$)

‡Significantly higher than 3 and 6 min ($P < 0.05$)

Neuromuscular function

The absolute values for neuromuscular parameters are reported in Table 2, while relative values (% of recovery) are displayed in Fig. 2. Compared to baseline, MVC, VA, $Q_{tw,pot}$, Q_{tw10} and Q_{tw100} were all significantly reduced ($P < 0.05$) 1 min after $P6_1$ (-19 ± 4, -6 ± 2, -37 ± 11, -52 ± 16 and -32 ± 11%, respectively). There was no significant change for M-wave ($+7 \pm 12$%, $P = 0.12$). The MVC and $Q_{tw,pot}$ recovered partially from 1 to 2 min (both $P = 0.01$) and from 2 to 5 min ($P = 0.04$ and 0.01), but there was no further significant recovery from 5 to 14 min ($P = 0.27$ and 0.15, respectively). There was a partial recovery from 1 to 2 min for Q_{tw10} ($P = 0.01$), without further significant recovery ($P > 0.05$). The Q_{tw100} recovered partially from 1 to 2 min ($P = 0.01$), without differences between 2 and 5 min ($P = 0.08$), and then further recovered partially between 5 and 14 min ($P = 0.01$). The VA recovered quickly after $P6_1$, returning to baseline values after 2 min of recovery ($P = 0.31$).

The time constant of *W′* and neuromuscular fatigue recovery

There was no significant difference between the time constant of *W′* recovery and the time constant of MVC, $Q_{tw,pot}$, Q_{tw10} and Q_{tw100} recovery ($P = 0.94$, 0.84, 0.13 and 0.23, respectively, Fig. 3). The time constant of VA recovery was significantly faster than the time constant of *W′* recovery ($P = 0.03$) and the time constant of Q_{tw100} recovery ($P = 0.01$), but was not significantly different from the time constant of MVC, $Q_{tw,pot}$ and Q_{tw10} recovery ($P = 0.21$, 0.34 and 0.98, respectively, Fig. 3). The time constant of Q_{tw100} recovery was significantly slower than the time constant of MVC, $Q_{tw,pot}$ and Q_{tw10} recovery ($P = 0.03$, 0.01, 0.01, respectively, Fig. 3).

Correlations between the time constant of W′ and neuromuscular fatigue recovery

There was a significant correlation only between the time constant of *W′* recovery and the time constant of MVC recovery ($r = 0.73$, $P = 0.02$, Fig. 4). However, there was no significant correlation between the time constant of *W′* recovery and the time constant of $Q_{tw,pot}$ ($r = -0.22$, $P = 0.53$), Q_{tw10} ($r = -0.13$, $P = 0.73$), Q_{tw100} ($r = 0.03$, $P = 0.95$) and VA ($r = -0.26$, $P = 0.47$, Fig. 4). There was also no significant correlation between the time constant of *W'* recovery and CP ($r = -0.44$, $P = 0.21$).

Correlations between neuromuscular fatigue after $P6_1$ and $P6_2$ and W′

The correlations between *W′* and changes in neuromuscular function from baseline to post-$P6_1$ were not significant (Table 3). Similarly, the correlations between *W′* and changes in neuromuscular function from baseline to post-$P6_2$ as well as between *W′* and changes in neuromuscular function from pre- to post-$P6_2$ were not significant, except a significant correlation ($r = 0.66$, $P = 0.04$) between *W′* and pre- to post-$P6_2$ change in VA (Table 3).

Training effect

The time to task failure during $P6_1$ was not different between the first and last experimental session (384 ± 44 and 386 ± 41 s, $P = 0.92$, respectively), suggesting no training effect due to successive exercise sessions.

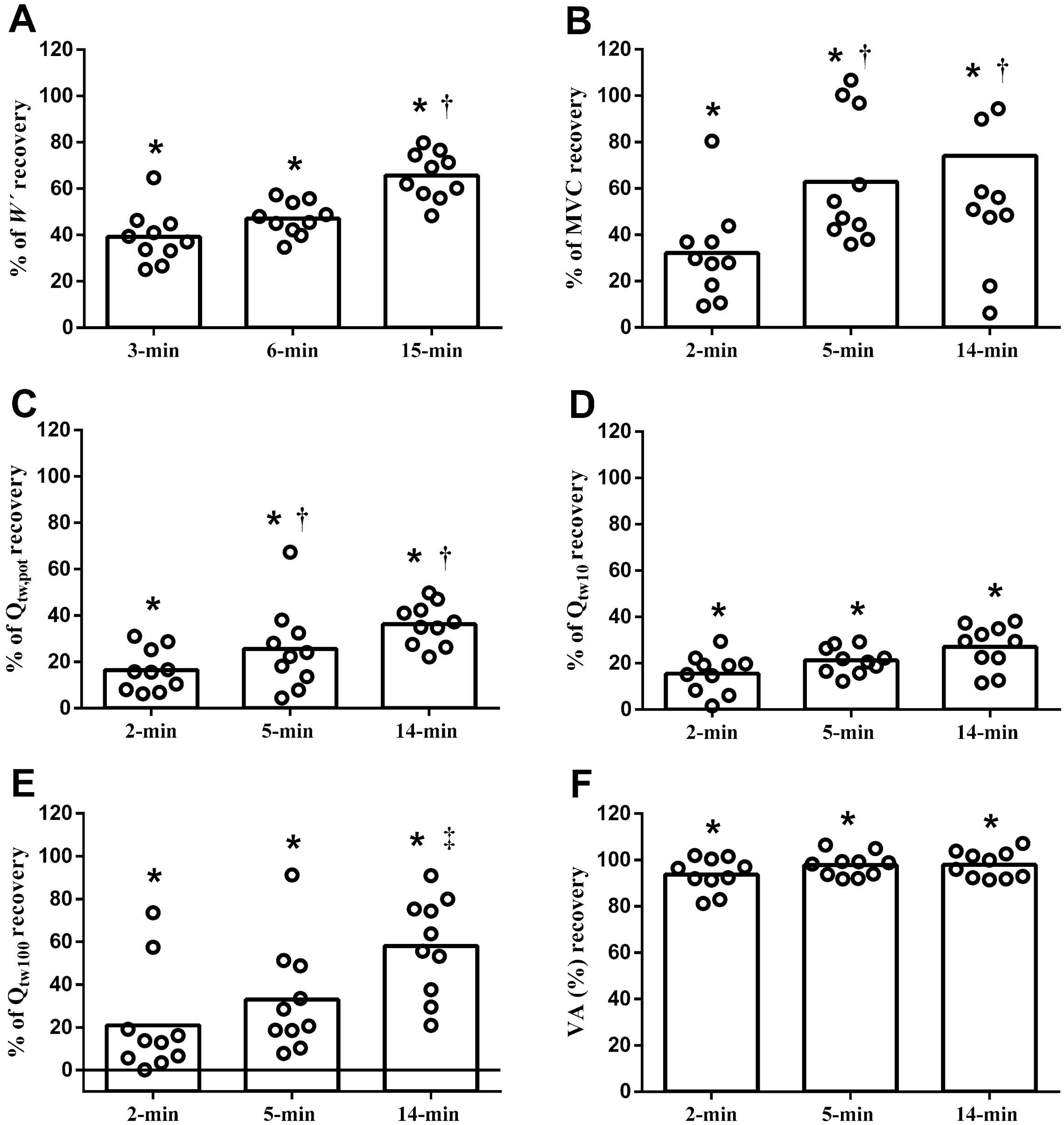

Fig. 2 Reconstitution of the capacity to work above critical power (W', **a**), maximal isometric voluntary contraction (MVC; **b**), potentiated quadriceps twitch torque evoked by single ($Q_{tw,pot}$; **c**), paired pulses (Q_{tw10} and Q_{tw100}; **d** and **e**, respectively) and voluntary activation (VA; **f**). After a trial designed to carry to task failure within 6 min ($P6_1$). Data are mean and individual plots ($n = 10$). * Significantly higher than $P6_1$ end ($P < 0.05$). † Significantly higher than 2-min recovery (or 3 min in Panel A) ($P < 0.05$). ‡ Significantly higher than 2- and 5-min recovery ($P < 0.05$)

Discussion

In the present study, the marker of central fatigue (VA) recovered faster than the W', but there were no significant differences between the time constant of W' recovery and the time constant of recovery of peripheral parameters ($Q_{tw,pot}$, $Q_{tw10,}$ and Q_{tw100}). However, the time constant of W' recovery was not associated neither with the time constant of peripheral nor with the time constant of central fatigue recovery. Rather, the time constant of W' recovery and the time constant of MVC recovery (a marker of overall neuromuscular fatigue) were not different and significantly associated with each other. These findings indicate that the recovery of W' does not depend exclusively on peripheral or central factors, but rather seems to be dependent on an overall process of recovery of neuromuscular function. Because MVC is an integrative expression of central and peripheral fatigue, these results suggest that both central and peripheral fatigue contribute jointly to W' recovery.

Table 2 Absolute values for neuromuscular function at baseline, and at 1, 2, 5 and 14 min after the $P6_1$

	Baseline	Post-exercise			
		1 min	2 min	5 min	14 min
MVC (N)	586 ± 160	441 ± 141*	472 ± 153*†	515 ± 139*†‡	521 ± 118*†‡
$Q_{tw,pot}$ (N)	222 ± 32	100 ± 29*	117 ± 11*†	135 ± 45*†‡	147 ± 27*†‡
Q_{tw10} (N)	315 ± 50	156 ± 45*	186 ± 46*†	197 ± 40*†	206 ± 39*†
Q_{tw100} (N)	296 ± 39	184 ± 49*	219 ± 56*†	226 ± 48*†	260 ± 28*†
M-wave (mV)	21 ± 7	23 ± 7	21 ± 9	22 ± 9	22 ± 8
VA (%)	91 ± 4	86 ± 9*	89 ± 12	90 ± 6	90 ± 5

Values are presented as mean ± SD (n = 10). MVC, maximal isometric voluntary contraction, $Q_{tw,pot}$, potentiated quadriceps twitch torque evoked by single pulse; Q_{tw10} and Q_{tw100}, paired pulses at 10 Hz and 100 Hz, respectively; M-wave, maximal M-wave amplitude and VA, voluntary activation

*Significantly lower than baseline ($P < 0.05$)

†Significantly higher than at 1 min ($P < 0.05$)

‡Significantly higher than at 2 min ($P < 0.05$)

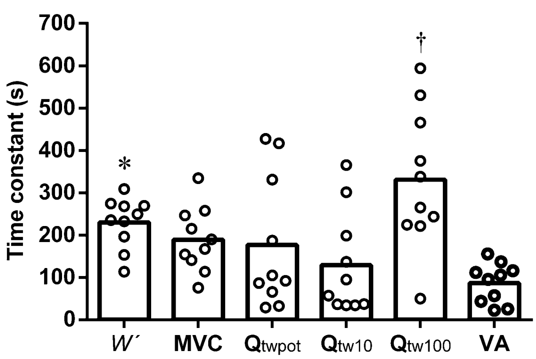

Fig. 3 Time constant of *W´* (capacity to work above critical power), maximal isometric voluntary contraction (MVC), potentiated quadriceps twitch torque evoked by single ($Q_{tw,pot}$), paired pulses (Q_{tw10} and Q_{tw100}) and voluntary activation (VA). Data are mean and individual plots (n = 10). * Significantly higher than VA. † Significantly higher than MVC, $Q_{tw,pot}$, Q_{tw10} and VA

The *W´* was partially restored within 3-min recovery, remained unchanged from 3- to 6-min recovery, and then had a further recovery within 15 min (Fig. 2a). However, the *W´* was not fully restored after 15 min of recovery (~ 66% of baseline). Previous studies found a higher recovery of *W´* in 15 min (~ 86% of baseline) (Ferguson et al. 2010). This lower recovery of *W´* in the present study might be explained by the lower level of aerobic fitness of our participants. The CP, an important parameter of aerobic fitness (Greco et al. 2012; Vanhatalo et al. 2016), was ~ 20–40% lower in our participants, compared to the CP previously reported (Ferguson et al. 2010; Skiba et al. 2012). In addition, our participants had a VO_{2max} ~ 30–45% lower than what has been previously reported (Ferguson et al. 2010; Skiba et al. 2012). The recovery of *W´* is partially related to the recovery of intramuscular PCr after an exhaustive exercise (Vanhatalo et al. 2016), and PCr recovery is dependent of aerobic metabolism (Layec et al. 2016). Endurance-trained individuals seem to have a rate of PCr resynthesis 27% faster than recreationally active individuals (Layec et al. 2016). Thus, less aerobically trained individuals will have a slower recovery of *W´*. In addition, we used a passive recovery between $P6_1$ and $P6_2$ because neuromuscular function had to be measured, while an active recovery (cycling at 20 W) was used in the aforementioned study (Ferguson et al. 2010), which may contribute to the lower recovery of *W´*.

The time constant of *W´* recovery was not different from the time constant of recovery of markers of peripheral fatigue (Fig. 3). However, the time constant of *W´* recovery was not significantly associated to the time constant of recovery of markers of peripheral fatigue (Fig. 4). The *W´* was also not associated to the magnitude of changes in peripheral fatigue after $P6_1$ or $P6_2$ (Table 3). In addition, while *W´* had already recovered ~ 66% after 15 min of recovery, markers of peripheral fatigue recovery varied highly, depending on the parameter analyzed (~ 36, 33% and 58% of baseline at 14 min of recovery for Q_{tw}, Q_{tw10} and Q_{tw100}, respectively, Fig. 2c, d and f, respectively). The initial recovery of *W´* (i.e., first 3 min) is likely related to the restoration of the energy systems and metabolic milieu (Ferguson et al. 2010). In fact, PCr (Cannon et al. 2014), extracellular potassium (Simmonds et al. 2010) and P_i (Kowalchuk et al. 2000) recover quickly (~ 5 min) after exercise and could influence the further ability to perform exercise above CP. On the other hand, intramuscular pH (Kowalchuk et al. 2000; Ravier et al. 2006) and sarcoplasmic reticulum Ca^{2+} uptake (Tupling et al. 2000) recover much slower (> 15 min), which will delay the full recovery of potentiated twitch force. Our findings indicate, therefore, that the processes involved in *W´* recovery might not be exclusively related to the processes involved in the recovery of peripheral fatigue.

The time constant of VA recovery was faster than the time constant of *W´* recovery (Fig. 3). Despite the predominance of peripheral fatigue in activities performed

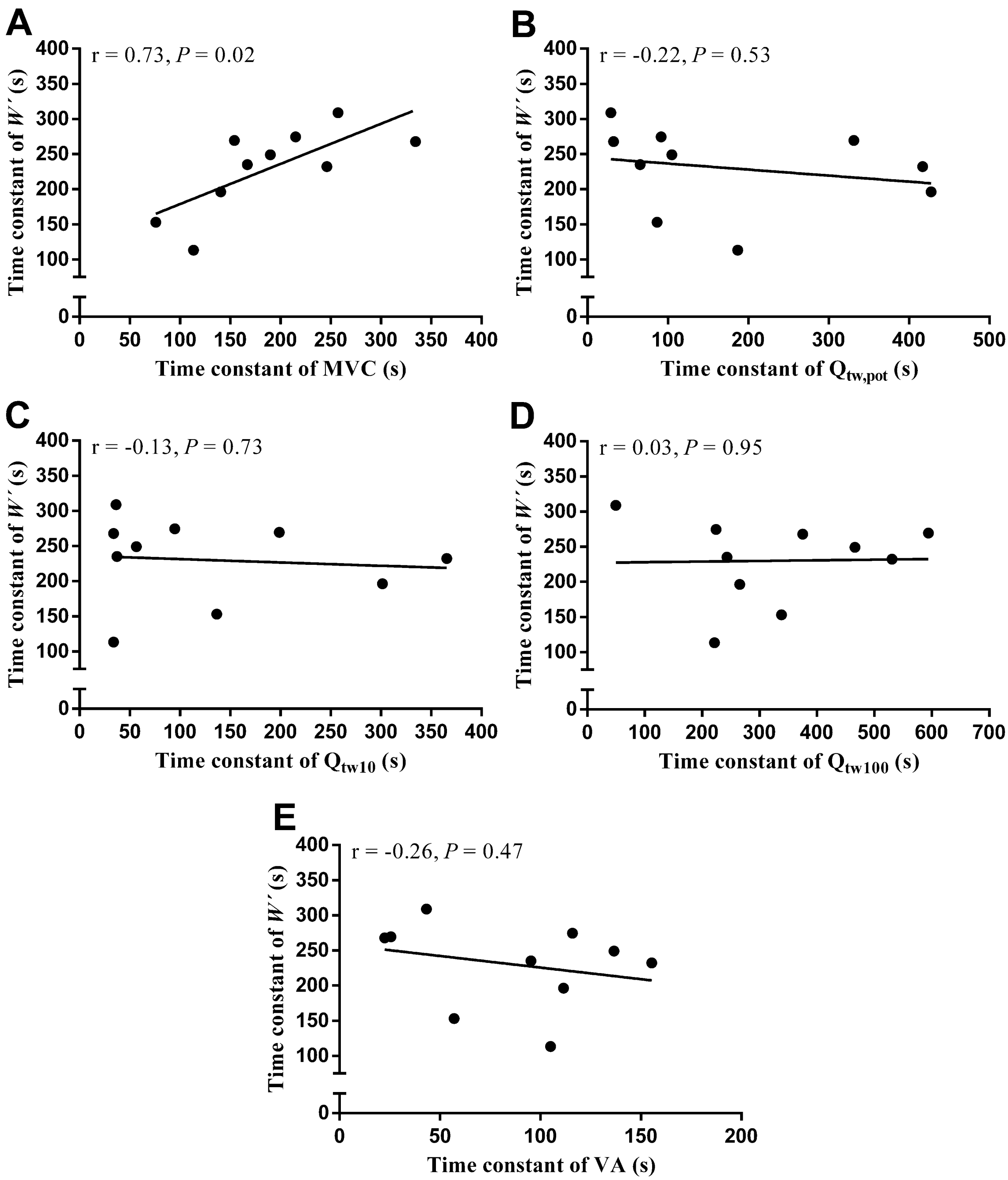

Fig. 4 Pearson coefficient of correlation ($n=10$) between the time constant of recovery of the capacity to work above critical power (*W*′) and the time constant of recovery of the maximal isometric voluntary contraction (**a**), potentiated quadriceps twitch torque evoked by single (**b**) and paired (**c**, **d**) pulses, and voluntary activation (**e**)

above CP, a significant occurrence of central fatigue has also been reported (Thomas et al. 2016). The full recovery of VA but not potentiated twitch force in the present study is in accordance with a study reporting that VA and potentiated twitch force decreases significantly after a Wingate test, but only the former is fully recovered within 30 min of passive rest (Fernandez-del-Olmo et al. 2013). It is noteworthy that performance during a second Wingate test, performed after this 30-min recovery, was not different from the first Wingate test (Fernandez-del-Olmo et al. 2013). These findings indicate that recovery of central fatigue may account for the restoration of performance after a prior high-intensity exercise performed above CP. However, despite this rapid recovery of central fatigue, the time constant of VA recovery was not significantly correlated to the time constant of *W*′ recovery in the present

Table 3 Pearson coefficient of correlation between the W' and changes in neuromuscular function from baseline to 1-min post-P6_1, from baseline to 1-min post each P6_2 trials (which was performed 3, 6 or 15 min after P6_1), and from immediately pre- to 1-min post each P6_2 trials

	% change (baseline to 1-min post-P6_1)	% change (baseline to 1-min post-P6_2)			% change (pre- to 1-min post-P6_2)		
		3 min	6 min	15 min	3 min	6 lmin	15 min
MVC	−0.08	−0.11	−0.15	−0.07	−0.07	−0.06	0.26
$Q_{tw,pot}$	0.37	0.21	0.09	0.09	−0.25	−0.24	0.13
Q_{tw10}	0.20	0.48	0.30	0.13	0.18	0.52	0.10
Q_{tw100}	0.30	0.63	0.31	0.29	0.13	0.49	0.39
VA	0.01	−0.41	0.55	−0.24	−0.15	0.66*	−0.29

MVC, maximal isometric voluntary contraction; $Q_{tw,pot}$, potentiated quadriceps twitch torque evoked by single pulse; Q_{tw10} and Q_{tw100}, paired pulses at 10 Hz and 100 Hz, respectively; VA, voluntary activation. * $P = 0.04$ ($n = 10$)

study. Together with a lack of correlation between recovery of peripheral fatigue and recovery of W', these results suggest that the recovery of W' does not depend solely from central or peripheral fatigue, but perhaps from a complex integration between peripheral and central systems, which may be represented by MVC.

Although there was a sharp decline in MVC after P6_1 (− 25%), which is comparable with the reduction reported in the literature after a similar exercise (Thomas et al. 2016), the MVC rapidly recovered, reaching ~ 74% of baseline within 14 min (Fig. 2). This faster recovery of MVC is in line with previous studies showing that the MVC recovers ~ 80% in the 4–5 min after a high-intensity exercise (Gandevia et al. 1996). It is also interesting to note that the recovery of MVC was time aligned and the only neuromuscular parameter significantly related to the recovery of W' (Fig. 4). The MVC and W' also shared similar characteristics, such as a fast recovery within the first few minutes after exercise, followed by a more gradual and slower recovery (Fig. 2). This is in agreement with previous findings showing similar behavior for recovery of MVC (Gandevia et al. 1996; Kennedy et al. 2015) and W' (Ferguson et al. 2010; Skiba et al. 2014). Interestingly, the capacity to develop maximal force depends on both central and peripheral systems (Todd et al. 2003). It has been demonstrated that during a sustained MVC protocol, ~ 25% of the total force reduction is caused by a decline in VA (Todd et al. 2003). It has also been demonstrated that, even maintaining a constant level of peripheral fatigue during repetitive concentric extension/flexion of the right knee, MVC declines parallel to a reduction in VA (Froyd et al. 2013). Although we found no significant relation between the time constant of VA and the time constant of MVC recoveries ($r = -0.15$, $P = 0.68$), it is noteworthy that VA had already fully recovered within 2 min after P6_1, anticipating a following recovery of MVC (Fig. 2). This rapid recovery of VA may have enabled a partial restoration of the capacity to generate maximal force, even though still with peripheral impairment. This partial resaturation of the capacity to generate maximal force coincided with the partial resaturation of the capacity to performing exercise above CP (i.e., to use the partially restored W').

Some limitations of the present study must be highlighted. Although effort was taken to start neuromuscular assessment after the P6_1 as soon as possible, there was a displacement delay from the cycle ergometer to the knee extension chair. Because there is some degree of both central (Gruet et al. 2014) and peripheral (Froyd et al. 2013) fatigue recovery within the 1st min after the exercise, our measurement of central and peripheral fatigue after the P6_1 may be slightly underestimated. However, studies performing neuromuscular assessment 1 or 2 min after a high-intensity exercise have been shown to be sensitive enough to detect significant central and peripheral fatigue (Blain et al. 2016; Vernillo et al. 2018). In addition, the neuromuscular fatigue was only measured in quadriceps muscle. During cycling exercise, many other muscle groups contribute to W' (e.g., hamstrings, calves and gluteal), but neuromuscular fatigue of these muscles was not measured, which may have contributed to weakening the relationship between W' and neuromuscular fatigue recoveries (Broxterman et al. 2019; Ferguson et al. 2010). Another limitation is that, as previously performed (Ferguson et al. 2010), we evaluated W' and fatigue recovery up to 15 min after exercise. A full recovery of W' and peripheral fatigue, however, could take hours or even days (Thomas et al. 2018). It would be interesting to have further studies to investigate the relationship between W' and peripheral fatigue over longer recovery periods. Likewise, we have recruited not trained participants, who would be susceptive to training effect because of the successive exercise sessions. However, there was no difference between the time to task failure in P6_1 between the first and last experimental trial, suggesting that a training effect may have minimally impacted our results. Together with reproducible values of maximal RPE across maximal incremental test and P6_1 and P6_2 trials, these findings also indicate that participants gave their maximal during all trials. In addition, although sample size calculation indicated that ten participants would be more than sufficient to detect differences in

W´ between the three recovery time points and sample size was similar to previous studies (Ferguson et al. 2010; Broxterman et al. 2016), some borderline *p* values (e.g., comparison of the time to task failure at $P6_2$ between 3- and 6-min recovery) suggest that a reduced statistical power cannot be fully disregarded. Finally, we adopted the inverse time model without testing whether other models would provide better determination of CP and *W´*. Nevertheless, the error of estimate for both CP and *W´* was close to those reported in the literature (Black et al. 2016; Schafer et al. 2019), suggesting minimal impact on our results.

Conclusion

The results of the present study revealed that the *W´* recovery is not exclusively dependent on the recovery of peripheral or central fatigue. Rather, the recovery of *W´* seems to be related to the recovery of the neuromuscular system's capacity to generate maximal force. Because the capacity to generate maximal force might depend of an integrative action of central and peripheral systems, our study suggests that recovery of *W´* may be anchored to an interplay between central and peripheral systems.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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